

Composition Determines Diversity: Categorical Fingerprints of Genetic Algorithms

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What is it?

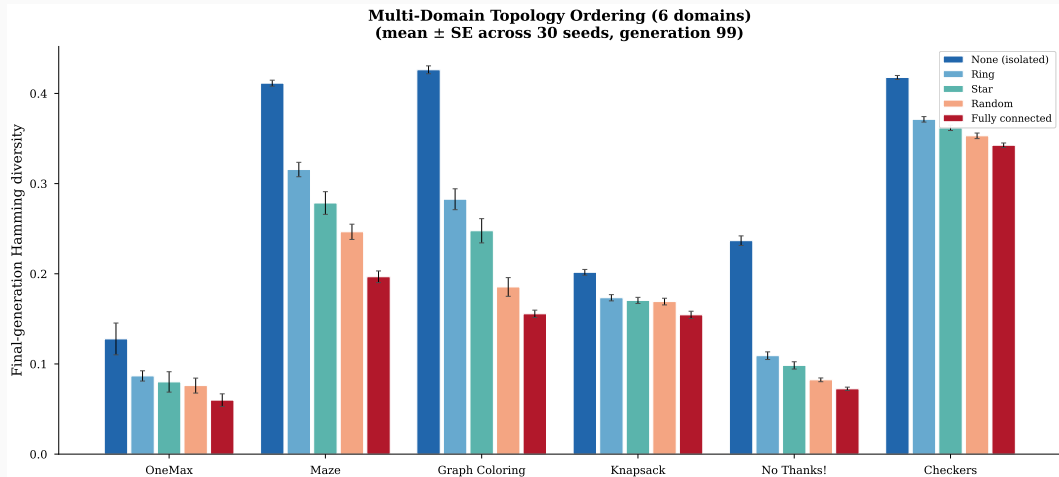
How different are the individuals in your population?

Measured as mean pairwise Hamming distance (normalized 0–1).

Why do we care?

- Too little → premature convergence, stuck in local optima
- Too much → no selection pressure, random search
- The sweet spot depends on the problem — but **what controls it?**

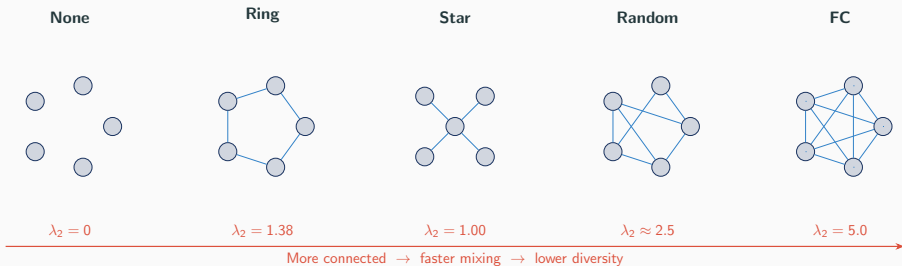
Multi-Domain Topology Ordering



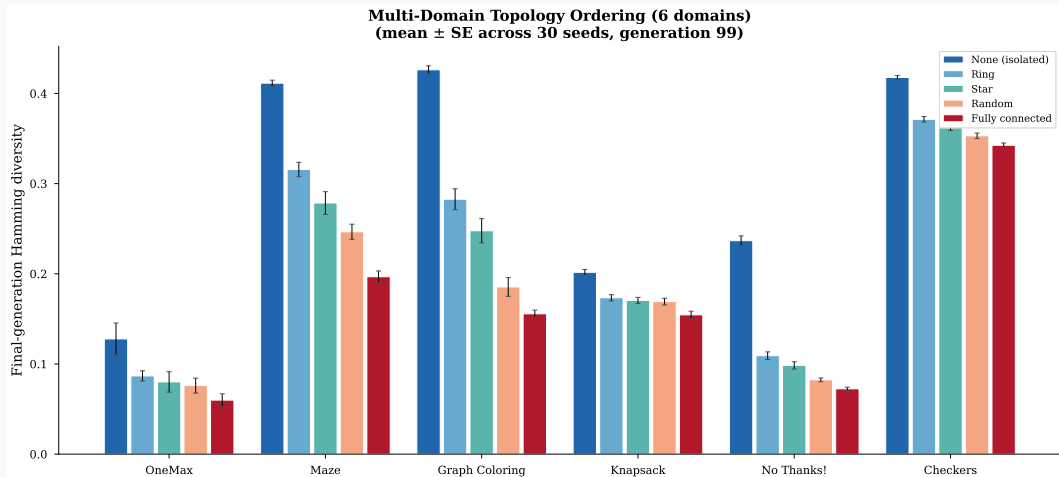
The Domains

1. OneMax
2. Maze (breed difficult perfect mazes — spanning trees on a grid — via edge-swap mutation & union crossover)
3. Graph Coloring
4. Knapsack
5. No Thanks!
6. Checkers

Island Models & Topology



Multi-Domain Topology Ordering



Kendall's W

$$W = 1.0$$

Perfect concordance

$$\chi^2 = 24.0, \quad p = 0.00008$$

Spearman

All 15 pairwise $\rho = 1.0$

Two-way ANOVA

$$F_{\text{topo}} = 47.8, \quad p < 10^{-6}$$

$$F_{\text{domain}} = 0.13, \quad p = 0.945$$

Variance ratio

$$23.9\times$$

Topology $\gg$ domain

The End

Nobody likes a long talk,
so you can tune out here if you like.

Up next: how Haskell and category theory
led us to this result.

Kleisli Arrows

Every GA operator has the same type:

$$\text{Pop } a \longrightarrow M(\text{Pop } a)$$

Population in, population out, effects bundled in M :

```
type EvoM = ReaderT GAConfig
            (WriterT GLog
             (State StdGen))
```

Kleisli composition ($\gg>$) chains them:

```
pipeline = evaluate >=> select
           >=> crossover
```

Why?

All code to reproduce these experiments:

<https://github.com/lyra-claude/categorical-evolution>